



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

**SECP3106-01 – APPLICATION DEVELOPMENT (WBL)**

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**LAB 2:**

**INDUSTRY PROJECT PROPOSAL**

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**PROJECT NAME:**

E-DUCATOR

**LECTURER NAME:**

DR AHMAD NAJMI BIN AMERHAIDER NUAR

**PREPARED BY: AD2Go**

| NAME                               | MATRIC NO |
|------------------------------------|-----------|
| NURUL IKA SYAFINY BINTI AZHAR      | A23CS0164 |
| LIM YU HAN                         | A23CS0241 |
| AIN NURNABILA BINTI MOHD AZHAR     | A23CS0207 |
| MOHAMED ALIF FATHI BIN ABDUL LATIF | A23CS0112 |
| NURUL ADRIANA BINTI KAMAL JEFRI    | A23CS0258 |

## Project Background

The current attendance process in the institution is not supported by a dedicated and unified system. Lecturers typically record attendance using external tools, which is not directly connected to timetable data or student records. As a result, tasks such as identifying absent students, checking class schedules, and maintaining consistent records require additional manual effort. This becomes more challenging when handling large numbers of students and multiple classes across different time slots. Beyond basic attendance marking, timetable management is also handled separately, making it difficult for lecturers to quickly access their schedule or navigate between classes. In cases where classes need to be rescheduled, the process lacks visibility and requires manual coordination.

To address these gaps, this project proposed the development of an integrated Attendance Management System that combines attendance recording, timetable handling, reporting, and class rescheduling within a single platform. This approach allows lecturers to manage their teaching sessions more efficiently while providing administrators with better control over scheduling and attendance data.

## Target Users

| User Role     | Description                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecturer      | <ul style="list-style-type: none"><li>- Record student attendance during class sessions using a checklist.</li><li>- View personal teaching timetable and access specific class sessions.</li><li>- Report student discipline issues such as frequent absences.</li><li>- Review attendance summaries and manage replacement classes when rescheduling is required.</li></ul> |
| Administrator | <ul style="list-style-type: none"><li>- Upload and manage timetable data, including class schedules, subjects, and lecturer assignments.</li><li>- Maintain accurate system records and ensure data consistency.</li><li>- Manage classroom availability to support class rescheduling and booking.</li><li>- Monitor attendance reports for institutional use.</li></ul>     |

Table 1: User Role and Description

## Mandatory Modules

This part shows each group member who is in charge of the Mandatory Modules, regardless of the industry project. They are considered core technical requirements of the subject and were intentionally excluded from the industry's six-module list.

| Module          | Module Leader                      |
|-----------------|------------------------------------|
| Login           | Mohamed Alif Fathi bin Abdul Latif |
| Registration    | Ain Nurnabila binti Mohd Azhar     |
| User Management | Nurul Ika Syafiny binti Azhar      |

Table 2: Mandatory Modules and Each Leader

## Six Industry Modules

Project E-ducator six industry-specific modules are aimed at solving the critical synchronization and data integrity problems that Dr. Osman is currently experiencing. Replacing a fragmented workflow between disparate platforms specifically, Microsoft Forms, Power Automate, Power BI, Canva code, and OneNote, these modules will ensure a smooth data flow between attendance records, student portal, and academic assessment. <sup>[OBJ]</sup>

M1: Taking Attendance is the main interface through which lecturers would record the attendance of students during live classes, which will allow the lecturers to precisely categorize the student status as present, absent, MC or CK.

M2: Report Discipline Issue offers lecturers a special purpose area to record instances of misbehaviour, or frequent absenteeism in students, which will support early intervention and ensure a reliable digital audit trail.

M3: Reporting Module is the centralized analytical engine of the system, in which all modules report their data, which is aggregated to generate real time attendance statistics and trends, including critical attendance percentages used to determine eligibility to take the exams by over 800 students.

M4: Upload Time Schedule is a feature that allows administrators to bulk-add timetable data such as class slots, subjects and lecturer assignments, and establish core data consistency at the beginning of each academic cycle.

M5: Displaying Timetable Slot demonstrates each lecturer as a teaching schedule in the form of a dashboard, where it is possible to find quick access to classes sessions and make a quick transition to attendance recording.

M6: Booking Module provides flexibility in the scheduling process, by allowing lecturers to reschedule classes and book replacement classes, replacing manual coordination with a transparent digital booking system.

### Module Distribution Table

| Module                            | Type      | Module Leader                      |
|-----------------------------------|-----------|------------------------------------|
| Login                             | Mandatory | Mohamed Alif Fathi bin Abdul Latif |
| Registration                      | Mandatory | Ain Nurnabila binti Mohd Azhar     |
| User Management                   | Mandatory | Nurul Ika Syafiny binti Azhar      |
| Module 1: Taking Attendance       | Industry  | Nurul Adriana binti Kamal Jefri    |
| Module 2: Report Discipline Issue | Industry  | Mohamed Alif Fathi bin Abdul Latif |
| Module 3: Reporting Module        | Industry  | Lim Yu Han                         |
| Module 4: Upload Time Schedule    | Industry  | Ain Nurnabila binti Mohd Azhar     |
| Module 5: Showing Timetable Slot  | Industry  | Nurul Adriana binti Kamal Jefri    |
| Module 6: Booking Module          | Industry  | Nurul Ika Syafiny binti Azhar      |

Table 3: Mandatory and Industry Modules with Each Module Leader

### Justification

The module distribution for Project E-ducator was strategically decided to balance technical complexity while maintaining a logical flow of data based on the industry partner's specific requirements. We grouped M1 (Taking Attendance) and M5 (Showing Timetable Slots) under Adlina to ensure a seamless transition from viewing a schedule to recording attendance. Similarly, Ika manages M6 (Booking Module) alongside Manage User, as both involve administrative control over resources and permissions. Alif handles M2 (Report Discipline Issue) and Login to ensure sensitive conduct records are protected by secure authentication, while Nabila manages M4 (Upload Time Schedule) and Registration, pairing the two primary "setup" functions of the system to ensure data consistency from the start.

The decision to assign Yuhan only one module, M3 (Reporting Module), is based on the significant technical depth required to solve the partner's previous data failures. This module must act as a centralized engine, aggregating data from all other modules to generate real-time statistics and trends, such as the critical 800+ student attendance percentages used to determine exam eligibility. By giving Yuhan a single, high-complexity focus, the team can ensure robust data synchronization and history tracking, addressing the specific "recurrent" data problems and Power BI connection issues highlighted in the briefing. This split ensures a fair workload distribution by balancing multiple straightforward tasks against one deep, analytical development challenge.

## **Conclusion**

By the end of the semester, our team aims to deliver a fully integrated web application that successfully transitions the industry partner's manual scheduling and fragmented tracking methods into a unified digital ecosystem. We aim to solve the critical synchronization issues previously experienced with Power BI by providing a robust reporting engine and a streamlined timetable management system that facilitates quick access to attendance recording. Ultimately, Project E-ducator will provide academic staff with a reliable tool for monitoring student discipline and attendance trends, ensuring operational efficiency and data integrity for the entire institution.